

AS Level Mathematics (Edexcel IAL) — Formula Sheet

Edexcel International Advanced Level, WMA11/12 (P1, P2) + applied units. The Edexcel booklet gives some of these; memorise them anyway — speed is marks.

PURE P1 — ALGEBRA & FUNCTIONS

Laws of indices	$a^p a^q = a^{p+q}$; $a^p \div a^q = a^{p-q}$; $(a^p)^q = a^{pq}$; $a^0 = 1/a^p$
Surds	"(ab) = " a " b; rationalise by multiplying by the conjugate
Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; discriminant $b^2 - 4ac$: > 0 two solutions
Completing the square	$ax^2 + bx + c = a(x + b/2a)^2 + (c - b^2/4a)$
Simultaneous equations	linear + quadratic! substitute; expect 0, 1 or 2 solutions
Inequalities	critical values; flip sign when multiplying by a negative; sketch regions

P1 — COORDINATE GEOMETRY

Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1}$; parallel $m_1 = m_2$; perpendicular $m_1 m_2 = -1$
Line forms	$y - y_1 = m(x - x_1)$; $ax + by + c = 0$
Distance & midpoint	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$; midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$
Circle	$(x - a)^2 + (y - b)^2 = r^2$; complete the square to find centre

P1 — CALCULUS

Differentiation	$\frac{d}{dx}(x^n) = nx^{n-1}$
Second derivative	$f''(x) > 0$ minimum; $f''(x) < 0$ maximum
Tangents & normals	gradient at $x = f'(x)$; normal gradient = $-1/f'(x)$
Increasing/decreasing	$f'(x) > 0$ increasing; $f'(x) < 0$ decreasing

P1 — TRIGONOMETRY & EXPONENTIALS

Exact values	$\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$; cos pairs swapped;
Identity	$\tan^2 \theta = \sin^2 \theta / \cos^2 \theta$
Sine & cosine rules	$a/\sin A = b/\sin B = c/\sin C$; $c^2 = a^2 + b^2 - 2ab \cos C$; area
Arc & sector (radians)	$s = r\theta$; $A = \frac{1}{2}r^2\theta$ (θ in radians!)

PURE P2

Binomial expand	$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$ (valid $ x < 1$)
Sine rule ambiguous case	check $\sin(180^\circ - \theta) = \sin \theta$
Trig identities	$\sin^2 \theta + \cos^2 \theta = 1$; $1 + \tan^2 \theta = \sec^2 \theta$; $1 + \cot^2 \theta = \csc^2 \theta$
Further calculus	$\frac{d}{dx}(e^x) = e^x$; $\frac{d}{dx}(\ln x) = 1/x$; $\frac{d}{dx}(\sin x) = \cos x$;
Integration	$\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$); $\int 1/x dx = \ln x + c$
Trapezium rule	$A \approx \frac{h}{2} [y_0 + 2(y_1 + \dots + y_{n-1}) + y_n]$

STATISTICS S1 (applied)

Mean & variance	$\bar{x} = \sum x/n$; $\text{Var} = \sum x^2/n - \bar{x}^2$; $SD = \sqrt{\text{Var}}$
Coding	$y = (x - a)/b$; $\sum y = (\sum x - a)/b$; Var scales by $1/b^2$
Probability	$P(A \cap B) = P(A)P(B)$ if independent; $P(A B) = P(A)$
Normal distribution	$Z = (X - \mu)/\sigma$; use tables; symmetrical about 0

MECHANICS M1 (applied)

suvat	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \frac{1}{2}(u+v)t$
Newton's 2nd law	$F = ma$ (F in N, result of all forces)

Connected particles

treat system as one for acceleration; split for tension

Moments

clockwise = anticlockwise about a point (equilibrium)